FIITJEE

JEE(Advanced)-2018

ALL INDIA TEST SERIES

ANSWERS, HINTS & SOLUTIONS
FULL TEST – I
PAPER-2

Q. No.	PHYSICS	Q. No.	CHEMISTRY	Q. No.	MATHEMATICS
1.	**C**	24.	**C**	47.	**C**
2.	**A**	25.	**B**	48.	**B**
3.	**C**	26.	**C**	49.	**B**
4.	**C**	27.	**D**	50.	**D**
5.	**A**	28.	**C**	51.	**A**
6.	**D**	29.	**B**	52.	**C**
7.	**B**	30.	**A**	53.	**D**
8.	**B**	31.	**D**	54.	**D**
9.	**ABC**	32.	**ABC**	55.	**BC**
10.	**BD**	33.	**D**	56.	**ABCD**
11.	**ABCD**	34.	**ABCD**	57.	**AC**
12.	**ABC**	35.	**ACD**	58.	**AB**
13.	**A**	36.	**A**	59.	**D**
14.	**A**	37.	**A**	60.	**B**
15.	**B**	38.	**A**	61.	**A**
16.	**B**	39.	**B**	62.	**B**
17.	**C**	40.	**A**	63.	**A**
18.	**D**	41.	**D**	64.	**D**
19.	**4**	42.	**9**	65.	**1**
20.	**3**	43.	**9**	66.	**4**
21.	**4**	44.	**6**	67.	**8**
22.	**3**	45.	**7**	68.	**1**
23.	**2**	46.	**7**	69.	**7**

Physics

PART – I

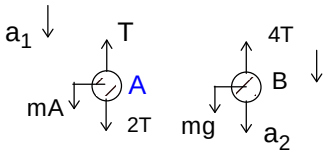
SECTION – A

1. For the 1st ball, $4h = \frac{gt_1^2}{2} \Rightarrow t_1 = 2\sqrt{\frac{2h}{g}}$ & For second ball $t_2 = \sqrt{\frac{2h}{g}}$

also $\sqrt{\frac{2h}{g}} + \sqrt{\frac{2d}{g}} = 2\sqrt{\frac{2h}{g}} \Rightarrow 2\sqrt{h} = \sqrt{d} + \sqrt{h} \Rightarrow d = h$

2. From F.B.D. we get

$$mA - 4T = mA_2$$
$$mA + 2T - T = mA_1$$



On solving these two we get

$$a_A = 4g$$

3. Zero error = +4 × 0.01 = +0.04 mm

Measured diameter = M.S.R. + L.C. × C.S.R.

= (5×0.5) mm + (0.01mm) × 18 = 2.68 mm

∴ Actual diameter = 2.68 - (0.04) = 2.64 mm

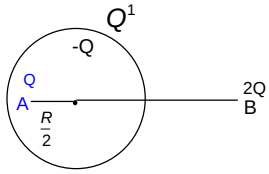
4. When radius of curvature is minimum $\bar{v}$ is $\perp$ to $\bar{a}$ & it is at t = $\frac{16V_0}{25g}$

5. Since shell is earthed its potential is zero

$$\frac{KQ^1}{R} + \frac{K(2Q)}{2R} = 0$$

$\therefore Q^1 = -Q$

Hence 3Q will flow to the earth.



6. $\frac{3}{2v_1} + \frac{1}{45} = \frac{1}{2R} \longrightarrow (1)$

$-\frac{3}{2v_1} = \frac{1}{6R} \longrightarrow (2)$

from (1) and (2)

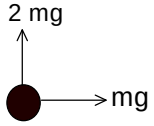
R = 30 cm

7. $F.v = k \Rightarrow .m\frac{dv}{dt}.v = k$

$\int_0^t dt = \frac{m}{k}\int_{v_i}^{v_t} v dv = \frac{m}{2k}\left(\frac{6M}{R} - \frac{6M}{r}\right)$

$t = \frac{6Mm}{2k}\left(\frac{1}{R} - \frac{1}{r}\right)$

8. F.B.D. of the ball in the frame of the container is as shown in the figure



$\therefore$ Distance from B = 5m

10. $\frac{Q_i^2}{2c}+\frac{1}{2}Li_i^2=\frac{1}{2}Li_{\max}^2$

$I_{\max}=\sqrt{2}\ amp$

Also, $\frac{Q_i^2}{2c}+\frac{1}{2}Li_i^2=\frac{Q_{\max}^2}{2c}$

11. $\because\ F=\frac{-du}{dy}=0$ (For equilibrium)

Particle is performing simple harmonic motion

12. $dQ=(4m)s(-dT_1)=(2m)(2s)dT_2$

On solving we get $\Delta T_f=\Delta T_i e^{\frac{-KAt}{2msl}}$

Temperature of point P remains constant

13. If x be the initial length of the gas chambers & A be the area of cross section of the cylinder then,

$x=\frac{V_0}{A}$ Also $P_0A = 2kx$

$\frac{P_0}{x}=2kx\Rightarrow P_0V_0=2kx^2$

$\therefore\frac{1}{2}k(2x)^2=\frac{1}{2}K4x^2=2kx^2=P_0V_0$

14. Similarly, $P_fA=2k\frac{V_f}{A}$

$P_0A=2k\frac{V_0}{A}$

$\Rightarrow\frac{P_f}{P_0}=\frac{V_f}{V_0}\Rightarrow P=\frac{P_0}{V_0}v$

Now $\therefore$ Q = dU + W

$=2\frac{5}{2}(P_1V_1-P_0V_0)+w$

$=12P_0V_0$

15 – 16.P= mv &

$\frac{P^2}{2m}=\frac{1}{2}mV^2$

$E_\alpha=\frac{P^2}{2m}$

$E_{radon}=\frac{P^2}{2M}$

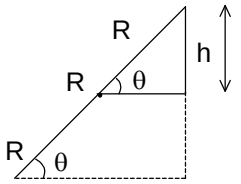
Also

$$\frac{P^2}{2m}+\frac{P^2}{2M}=\Delta E \Rightarrow P=\sqrt{2\Delta E\frac{mM}{m+M}}$$

$m=4.002mu, M=222.017mu$

$m=6.64\times10^{-27}kg$

17 – 18. $\frac{1}{2}g\sin\theta.\frac{4v^2}{g^2\cos^2\theta}=2R \quad .....(I)$



$v^2=Rg\sin\theta \quad .....(2)$

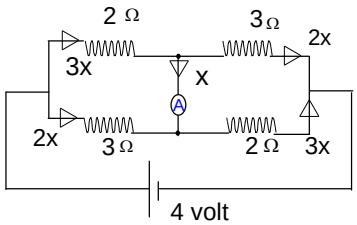
From (1) and (2) $\tan\theta=1$

SECTION – C

19. $v=\sqrt{\frac{T}{\sigma}}=\sqrt{\frac{\frac{2.5}{0.75}}{\frac{0.125}{0.75}}}=3.9m/s\simeq 4m/s$

20. ∵ Angular separation is directly proportional to wave length

21.



$2x=4$

$x=\frac{1}{3}=\frac{p}{q}$

$p+q=4$

22. $\because \tau=I\alpha \Rightarrow \omega=\sqrt{\frac{8QE}{\pi mR}} \Rightarrow \omega=\sqrt{2\pi}=\frac{2\pi}{T}$

$\therefore T=\sqrt{2\pi}$

∴ P + n is 3

23. Now $i=\frac{\varepsilon}{R}=\frac{\pi r^2}{R}\frac{dB}{dt}$

$\left|\vec{\tau}\right|=\left|\vec{M}\times\vec{B}\right|=2mgr$

$\frac{\pi r^2}{R}\frac{dB}{dt}.\pi r^2.4=2mg.r$

$4\frac{\pi^2 r^4}{R}.8t=2mg.r$

$t=2\sec$

Chemistry

PART – II

SECTION – A

24. $CHCl_3 \rightarrow \overline{C}Cl_3 + H^+$

$CHF_3 \rightarrow \overline{C}F_3 + H^+$

$\Rightarrow$ $\overline{C}Cl_3$ is more stable than $\overline{C}F_3$ due to $P_\pi - d_\pi$ bonding hence $CHCl_3$ is more acidic, not CHF_3 so, (I) is incorrect.

$\Rightarrow$ In chlorobenzene $C-Cl$ bond length decreases due to resonance and hence its dipole moment is lesser than chlorocyclohexane ii is also incorrect.

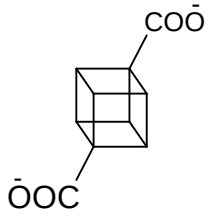
$\Rightarrow$ Dipole moment of fluoromethane is lesser than chloromethane due to very small size of F. ii is also incorrect.

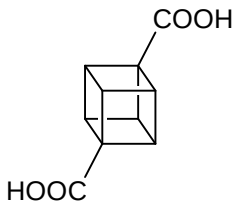
$\Rightarrow$ Carbon and nitrogen atom containing organic compound gives positive test for nitrogen in lassaigne's test

$\Rightarrow$ Dipole moment of difluoromethane is more than dichloromethane .

CH_2F_2	CH_2Cl_2
$\mu = 1.97D$	$1.60D$

25.

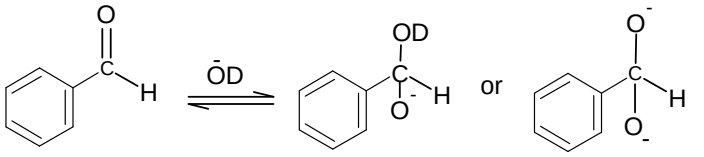
A. 

B. 

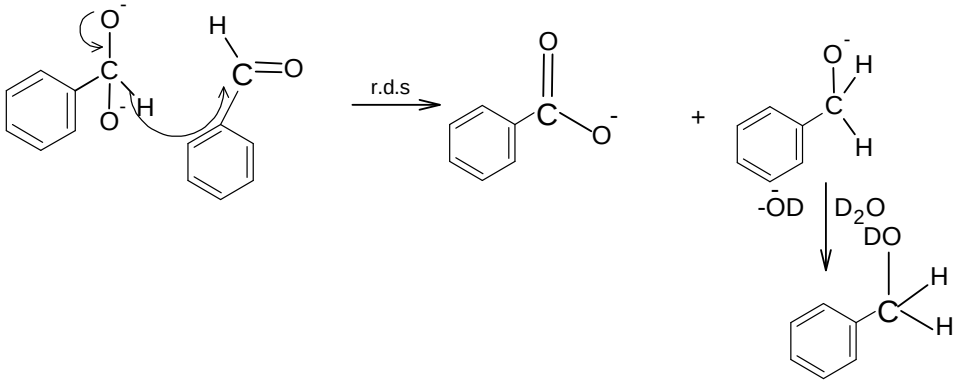
C. 

26. At very high concentration R. O. R = $K\left[RCHO\right]^2[OD^-]^2$

Mech.

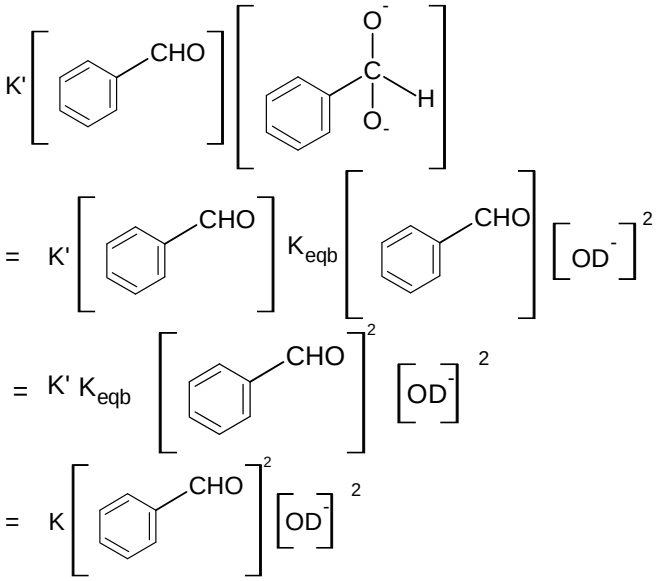


(At very high concentration of OD^-)



At very high concentration =

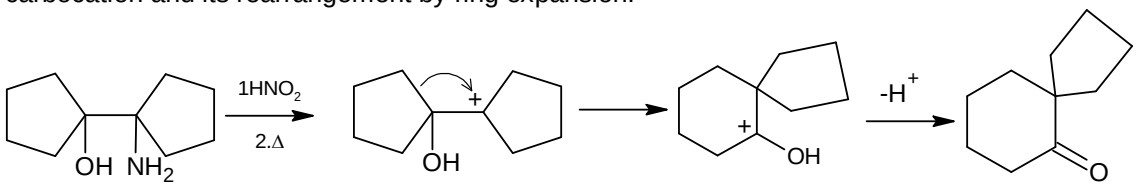
r. o. r =



27. $Ni(CO)_4$, $Fe(CO)_5$, $Cr(CO)_6$ ↓ Extent of synergic bonding ↓ M - C bond strength ↑ CO bond strength

$[V(CO)_6]^-$, $[Cr(CO)_6]$, $[Mn(CO)_6]^+$ ↑ Extent of synergic bonding ↑ M - C bond strength ↓ CO bond strength

28. It is a semi pinacol – pinacolone type reaction and product is formed after formation of carbocation and its rearrangement by ring expansion.



29. $2\pi r = 4\lambda; n = 4$

Total energy required + Total energy released = 0

$$(2\times 4.526ev\times N_A)+(2\times 13.6N_A)+\left(2\times 13.6\times\left(1-\frac{1}{16}\right)\times N_A\right)-(30.87\times x\times N_A)=0$$

$x = 2$

30. $AgBr \rightleftharpoons Ag^+_{(aq.)} + Br^-_{(aq.)}$

$K_{sp} = [Ag^+][Br^-]$

$Ag^+ + 2NH_3 \rightleftharpoons [Ag(NH_3)_2]^+$

$$K_f = \frac{\left[Ag\left(NH_3\right)_2\right]^+}{\left[Ag^+\right]\left[NH_3\right]^2} = 1\times10^8$$

Let s be the solubility of AgBr in NH_3, then $[Br^-] = [Ag^+] + [Ag(NH_3)_2]^+ = S$
Since , almost all Ag^+ in solution state passes to $[Ag(NH_3)_2]^+$ as K_f is very high. Thus $[Ag^+]<<<[Ag(NH_3)_2]^+$
$\therefore [Br^-] = [Ag(NH_3)_2]^+ = S$

Now $K_{sp} = [Ag^+][Br^-] = \frac{\left[\left[Ag(NH_3)_2\right]^+\right]}{K_f\times\left[NH_3\right]^2}\times\left[Br^-\right]$

Since $[NH_3] = 0.4$ M

$$5\times10^{-13} = \frac{S\times S}{1.0\times10^8\times(0.4)^2}$$

$$S^2 = 8\times10^{-6}$$

$$S = 2.82\times10^{-3}M$$

31. $Fe \rightarrow Fe^{+2} + 2e \qquad -\Delta G_1^0 = 2\times E_1^0\times F.....(i)$

$Fe^{+2} \rightarrow Fe^{+3} + e \qquad -\Delta G_2^0 = 1\times E_2^0\times F.....$(ii)

$Fe \rightarrow Fe^{3+} + 3e \qquad -\Delta G_3^0 = 3\times E_3^0\times F.....$(iii)

Subtracting equation (i) from equation (ii), we get

$Fe^{+2} \rightarrow Fe^{+3} + e$

$-\Delta G_3^0 + \Delta G_1^0 = 3E_3^0F - 2E_1^0F.....$(iv)

Comparing (ii) and (iv)

$-\Delta G_2^0 = -\Delta G_3^0 + \Delta G_1^0$

$E_2^0F = 3E_3^0F - 2E_1^0F$

$E_3^0 = \left(E_2^0 + 2E_1^0\right)/3$

32. Crystal is FCC

33. $2MnO_4^- + 5H_2O_2 + 6H^+ \rightarrow 2Mn^{+2} + 5O_2 + 8H_2O$

34. $O_{(g)} + e^- \rightarrow O_{(g)}^- \qquad \Delta H = -ve$

$O_g^- + e^- \rightarrow O_{(g)}^{-2} \qquad \Delta H = +ve$

$O_{(g)} + 2e^- \rightarrow O_{(g)}^{-2} \qquad \Delta H = +ve$

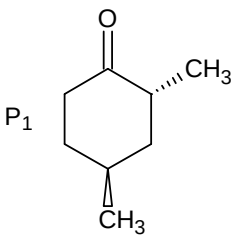
$O_{2(g)} \rightarrow O_{2(g)}^+ + e^- \qquad \left(Electron\ removed\ from\ \pi^*2P_y\ or\ \pi^*2P_x\right)$

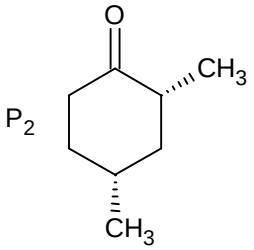
$O_{(g)} \rightarrow O_{(g)}^+ + e^- \qquad \left(Electron\ removed\ from\ \ 2P\right)$

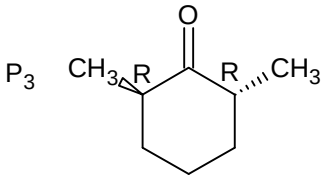
$H_{2(g)} \rightarrow H_{2(g)}^+ + e^- \qquad \left(Electron\ removed\ from\ \ \sigma1s\right)$

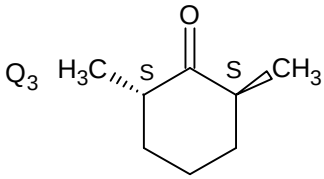
$H_{(g)} \rightarrow H_{(g)}^+ + e^- \qquad \left(Electron\ removed\ from\ \ 1s\right)$

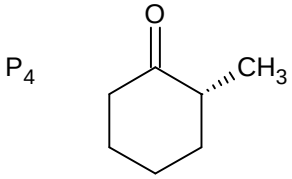
35.



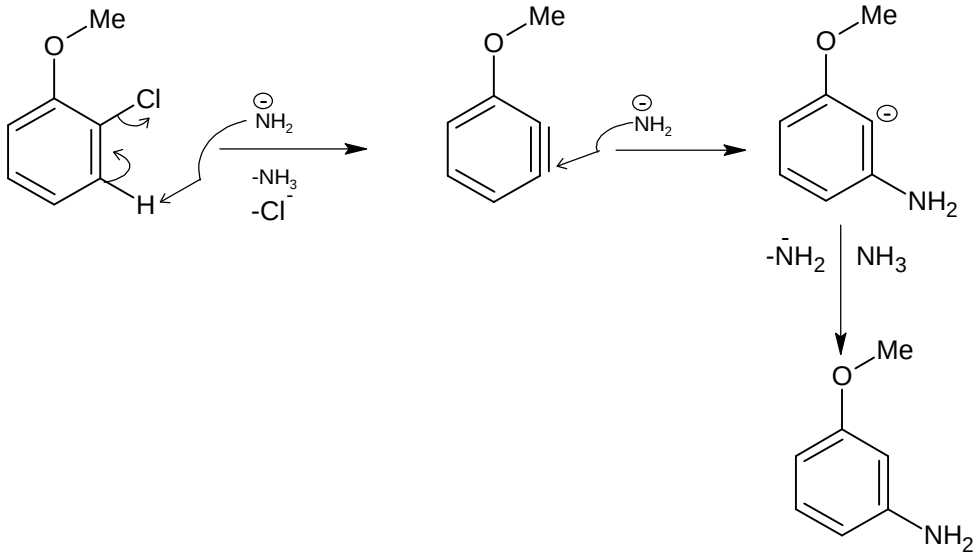

O
CH3
P2
CH3


O
P3
CH3
R
R
CH3

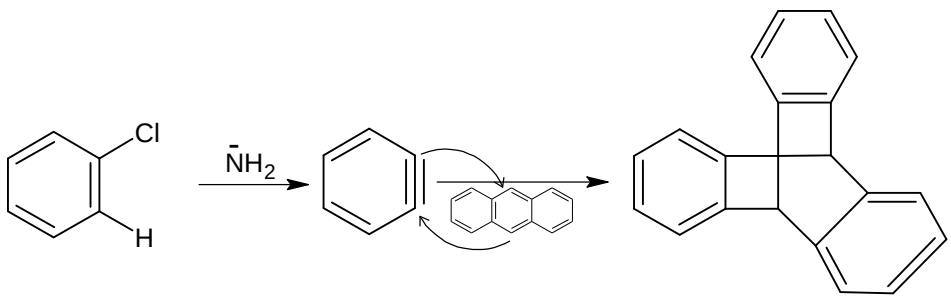

O
Q3
H3C
S
S
CH3


O
P4
CH3

36.


Me
O
Cl
H
NH2
-NH3
-Cl-
-NH2
NH3
NH2

37.


Cl
H
NH2

38. $\frac{PV}{nRT} = Z$ At very low P or P $\rightarrow$ 0, Z = 1

39. $PV = RT$ (for 1 mole of gas)

$\therefore P\partial V = R\partial T$

or $\left(\frac{\partial V}{\partial T}\right)_P = \frac{R}{P} = \frac{R.V}{RT}$

or $\alpha = \frac{\left(\frac{\partial V}{\partial T}\right)_P}{V} = \frac{1}{T}$

$P.\partial V + V.\partial P = 0$

$\left(\frac{\partial V}{\partial P}\right)_T = -\frac{V}{P}$

Also

or $\beta = -\frac{\left(\frac{\partial V}{\partial P}\right)_T}{V} = \frac{1}{P}$

$\frac{\alpha}{\beta} = \frac{P}{T}$

40-41. Alkaline gas C with NaOH indicates that solid should be ammonium salt. Heating the salt, a colorless non-reacting gas D is formed. The gas D may be nitrogen . The compound may thus be NH_4NO_2 . Reactions involved are as follows.

$NH_4NO_2 \xrightarrow{HCl} NH_4Cl + HNO_2$

A

$2HNO_2 \longrightarrow H_2O + 2NO + [O]$

$2NO + 2[O] \longrightarrow 2NO_2 \uparrow$

B

$Brown Gas$

$NH_4NO_2 + NaOH \longrightarrow NaNO_2 + H_2O + NH_3 \uparrow$

C

$NH_4NO_{2(s)} \xrightarrow{heat} N_{2(gas)} \uparrow + 2H_2O_{(gas)} \uparrow$

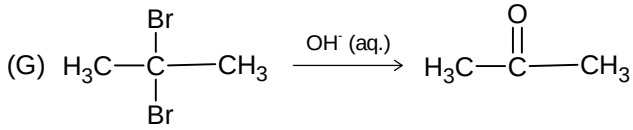
A D

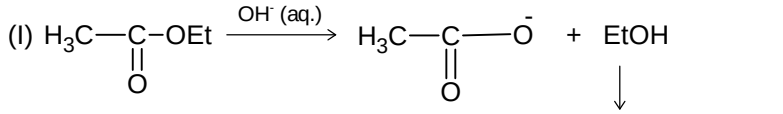
$\Rightarrow N_2$ *is a diamagnetic Gas*

$\Rightarrow$ *Bond order of* $N_2 = 3$

SECTION – C

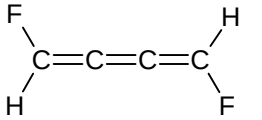
42. (A), (B), (C), (D), (E), (F), (G), (H) & (I) gives iodoform test

(G) 

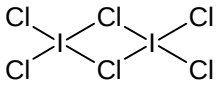
(I) 

it show iodoform test

43. G & J have zero dipole moment



$\mu = 0$



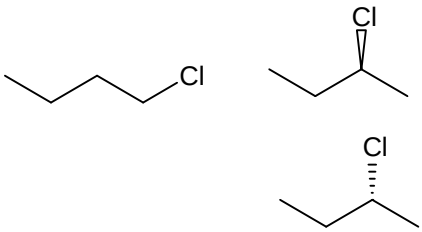
$\mu = 0$

A, B, C, D, E, F, H, I have nonzero dipole moment

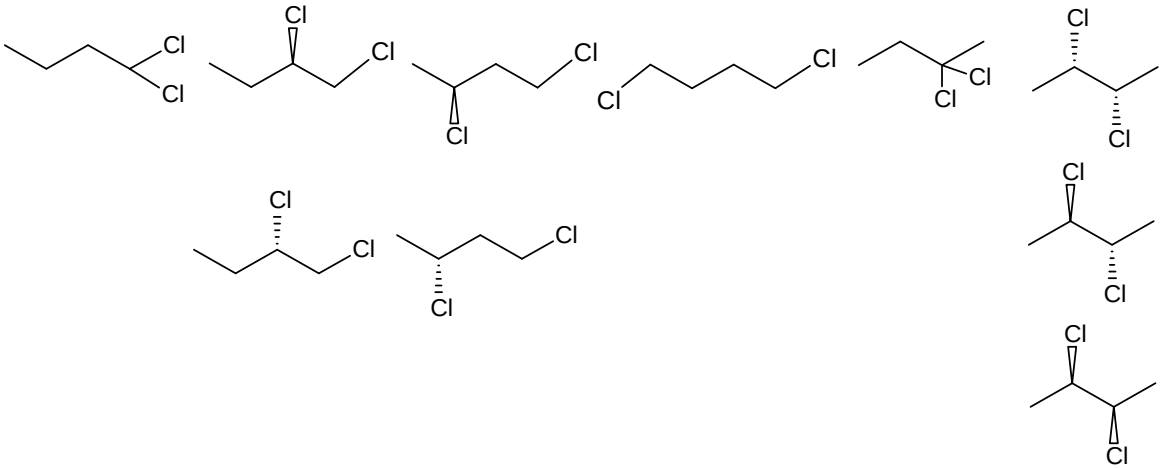
44. Correct statements: A, C, D, G, H, I
Incorrect statements: B, E, F

45. A, B, C, D, G, H, I show mutarotation. A sugar whose IUPAC name ends with the suffix ose is a reducing sugar and they do mutarotation

46. Monochlorination



Dichlorination



A, B, C, D, E, F, J are correct.

Mathematics

PART – IIII

SECTION – A

47. $S_\eta = 1+\frac{1+3}{2!}+\frac{1+3+3^2}{3!}\ldots\ldots\ldots\ldots\ldots = \sum\frac{3^n-1}{(3-1)n!}$

$\frac{1}{2}\sum_{r=1}^{n}\frac{3^r-1}{r!} = \frac{1}{2}\left[\sum\frac{3^r}{r!}-\frac{1}{r!}\right]$

$\lim_{n\to\infty}(S_n) = \frac{1}{2}\left[\left(e^3-1\right)-(e-1)\right] = \frac{1}{2}\left[e^3-e\right] \approx 8.59$

$[S_n] = 8$

48. Let $f(x) = a(x-2)^4 + b(x-2)^3 + C(x-2)^2 + d(x-2) - 1$

$f(x) = 12(x-2)^2 - 12(x-2) + 2$

$f''(1) = 26$

49. Let E_n denote the event that the score is n

$E_n = (E_{n-2} \cap H) \cup (E_{n-1} \cap T)$

$P(E_n) = \frac{1}{2}[P_{n-2} + P_{n-1}]$

$P_n + \frac{1}{2}P_{n-1} = P_{n-1} + \frac{1}{2}P_{n-2}$

$= P_{n-2} + \frac{1}{2}P_{n-3} = P_2 + \frac{1}{2}P_1$

As, $P_1 = P(T) = \frac{1}{2} \Rightarrow P_2 = P((T \cap T) \cup H) = \frac{3}{4}$

$\Rightarrow P_n + \frac{1}{2}P_{n-1} = 1$

$\Rightarrow P_n - \frac{2}{3} = -\frac{1}{2}\left[P_{n-1} - \frac{2}{3}\right] = \left(-\frac{1}{2}\right)^2\left[P_{n-2} - \frac{2}{3}\right]$

$= \left(-\frac{1}{2}\right)^{n-2}\left[P_1 - \frac{2}{3}\right] = -\frac{1}{2}\left[\frac{1}{2} - \frac{2}{3}\right]$

$P_{100} = \frac{2}{3} + \frac{1}{3.2^{100}}$

50. $x - 1 < [x] \le x$

$\frac{1}{\eta^3} - 1 < \left[\frac{1}{\eta^3}\right] \le \frac{1}{\eta^3}$

$x^5\left(\frac{1}{\eta^3} - 1\right) < x^5\left[\frac{1}{x^3}\right] \le x^2$

$\Rightarrow \lim_{x\to 0} x^5\left[\frac{1}{x^3}\right] = 0$

51. Required number of ways = coefficiant of x^{150} in

$\left(x^0+x+x^2.....x^{50}\right)^3\ \left(x^0+x+x^2.....x^{100}\right)$

$^{153}c_{150}-3\ ^{102}c_{99}-^{52}c_{49}+3.^{51}c_{48}$

$=70176+40375=110551$

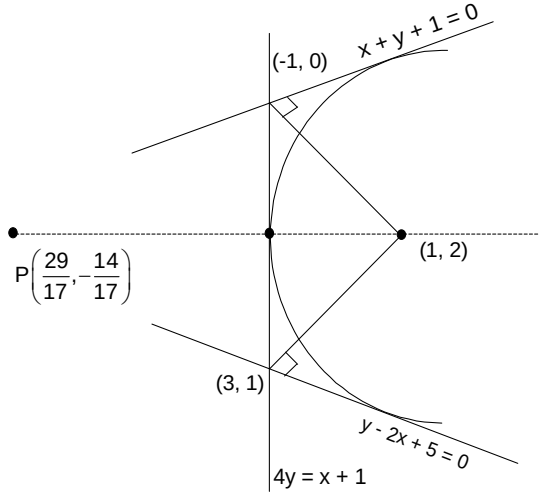
52. $e^{xy}\left[xy^2+y^3\dfrac{dx}{dy}\right]+e^{\frac{x}{y}}\left[y\dfrac{dx}{dy}-x\right]=0$

$d\left(e^{xy}\right)+d\left(e^{\frac{x}{y}}\right)=0$

$e^{xy}+e^{\frac{x}{y}}+c=0$

$A+B=1$

53. Normal through given point be the axis of parabola,
y – 2= - 4(x – 1)
4x + y = 6



54. Let a, b, c are the roots of equation

$a_0x^3+a_1x^2+a_2x+a_3=0$

Then, $s_1=5 \quad s_2=12, \quad s_3=25$

Now, $a_0s_1+a_1=0 \quad \Rightarrow a_1=-5a_0$

$a_0s_2+a_1s_1+2a_2=0\Rightarrow a_2=\dfrac{13}{2}a_0$

$a_0s_3+a_1s_2+a_2s_1+3a_3=0\Rightarrow a_3=\dfrac{5}{6}a_0$

$x^3-5x^2+\dfrac{13}{2}x+\dfrac{5}{6}=0$

$6x^3-30x^2+39x+5=0$

$\Rightarrow 6x^4-30x^3+39x^2+5x=0$

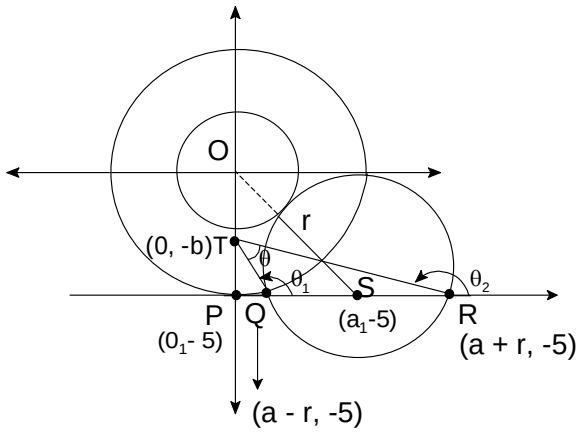
$6s_4-30s_3+39s_2+5s_1=0\Rightarrow s_4=\dfrac{257}{6}$

55. $\tan\theta = \dfrac{\dfrac{-5+b}{a+r} - \dfrac{-5+b}{a-r}}{1-\dfrac{(b+5)^2}{a^2-r^2}}$

Now, $a^2 = (r+4)^2 - 5^2$

$= r^2 - 9 + 8r$

$(b-5)^2 = 9$



56. $\int_0^x g(t)\,dt = 2 - K^2x^2 = 2 - \int_0^x 2K^2 t\,dt$

Let $f(x) = \int_0^x \left[2k^2t + g(t)\right]dt - 2$

As, $2K^2t$ and $g(t)$ are continuous

$\Rightarrow \left[2K^2t + g(t)\right]$ is also continuous

$f(0) = -2$, $\lim_{x\to\infty} f(x) \to \infty$

As $f(x)$ changes its sign

$\Rightarrow f(x) = 0$ for some $x \in R$ & $\forall K \in R$

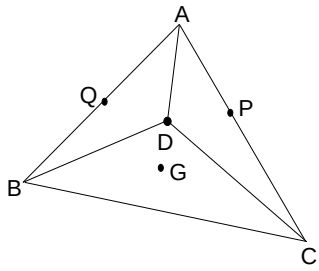
57. $\overrightarrow{DA} = \vec{a}, \overrightarrow{DB} = \vec{b}, \overrightarrow{DC} = \vec{c}$

$\overrightarrow{AD}.\overrightarrow{BC} = (-\vec{a}).(\vec{c} - \vec{b})$

$= -\vec{a}.\vec{c} + \vec{a}.\vec{b}$

$= 0$

Hence, $\overrightarrow{AD} \perp^r \overrightarrow{BC} \Rightarrow \overrightarrow{AB} \perp \overrightarrow{CD}$



Now, $\overrightarrow{PG} = -\dfrac{1}{6}(3\vec{a} - 2\vec{b} + \vec{c})$ & $\overrightarrow{DQ} = \dfrac{1}{2}(\vec{a} + \vec{b})$

Let $|\vec{a}| = |\vec{b}| = |\vec{c}| = K$

$\Rightarrow |\overrightarrow{PG}| = \dfrac{K}{2}$ & $|\overrightarrow{DQ}| = \dfrac{\sqrt{3}}{2}K$

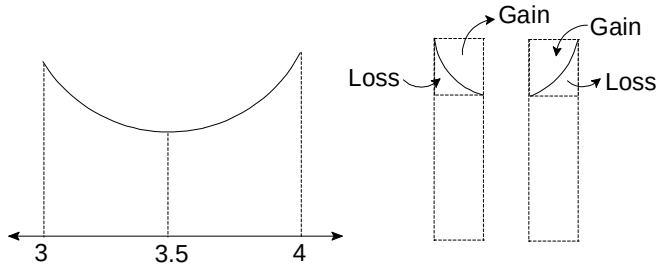
$\Rightarrow \overrightarrow{PG}.\overrightarrow{DQ} = -\dfrac{1}{6}(3\vec{a} - 2\vec{b} + \vec{c}).\dfrac{1}{2}(\vec{a} + \vec{b})$

$\cos\theta = -\dfrac{5}{6\sqrt{3}} \Rightarrow \theta = \pi - \cos^{-1}\left(\dfrac{5}{6\sqrt{3}}\right)$

58. $f(x) = x^2 - 7x + 13$

In both cases gain > loss

$\Rightarrow P > \int_3^4 f(x)dx \quad \& \quad Q > \int_3^4 f(x)dx$



59. & 60.

As, $f(x) = x$ & $\cos(n+1)\theta + \cos(n-1)\theta = 2\cos(n\theta).\cos\theta$

$f_{(n+1)}(x) + f_{(n-1)}(x) = 2x.f_n(x)$

$f_n(x) = \frac{1}{2x}\left[f_{n+1}(x) + f_{n-1}(x)\right]$

Now, Put $x = \cos\theta, \Rightarrow \sqrt{x^2-1} = i\sin\theta$

$\left(x + \sqrt{x^2-1}\right)^{10} + \left(x - \sqrt{x^2-1}\right)^{10} = (\cos\theta + i\sin\theta)^{10} + (\cos\theta - i\sin\theta)^{10}$

$= 2\cos(10\theta) = 2f_{10}(\cos\theta) = 2f_{10}(x)$

61. & 62.

(1) Put $(2x-3)^{\frac{1}{6}} = t$

(2) Put $\frac{x+1}{1-x} = t^3$

63. & 64.

equation of line L_2 is $\frac{x-3}{1} = \frac{y+4}{-3} = \frac{z-1}{-1}$

(1) equation of plane containing L_1 & L_2

$$\begin{vmatrix} x-3 & y+1 & z-2 \\ 0 & 3 & 1 \\ -1 & 6 & 2 \end{vmatrix} = 0 \Rightarrow y + 3z - 5 = 0$$

(2) volume of tetrahedron $= \frac{1}{6}\begin{vmatrix} 3 & -4 & 1 \\ 5 & -4 & 1 \\ 7 & -7 & 0 \end{vmatrix} = \frac{7}{3}$

SECTION – C

65. equation of chord whose mid-point is $(t, K-t)$

$\frac{tx}{8} + \frac{(K-t)y}{2} = \frac{t^2}{8} + \frac{(K-t)^2}{2}$ .....(i)

$(2,-1)$ satisfy (i)

$\Rightarrow 5t^2 - (6+8K)t + 4(K^2+K) = 0$

For two distinct chords,

$D > 0$

$(6+8K)^2 - 4.4.5(K^2+K) > 0$

$\Rightarrow 4K^2 - 4K - 9 < 0$

$K \in \left[\frac{1-\sqrt{10}}{2}, \frac{1+\sqrt{10}}{2}\right] \Rightarrow a+b=1$

66. Let $f(x) = ax^2 + \frac{b}{x} - c \qquad f'(x) = 2ax - \frac{b}{x^2}$

$f'(x) = 0, \quad x = \left(\frac{b}{2a}\right)^{\frac{1}{3}} \Rightarrow a\left(\frac{b}{2a}\right)^{\frac{2}{3}} + \frac{b}{\left(\frac{b}{2a}\right)^{\frac{1}{3}}} \geq c \quad \Rightarrow \frac{27ab^2}{c^3} \geq 4$

67. $x+y+z=1$

$(y+z)+(z+x)+(x+y)=2$

Let $y+z=A,\ z+x=B, \quad x+y=C$

$(1+x)(1+y)(1+z) = (B+C)(C+A)(A+B)$

$\Rightarrow (1+x)(1+y)(1+z) \geq 8(x+y)(y+z)(z+x) \qquad .....(i)$

As, $(x+y)(y+z)(z+x) = (1-x)(1-y)(1-z)$

$= 1-(x+y+z)+(xy+yz+zx)-xyz$

$= xy+yz+2x-xyz$

$(x+y)(y+z)(z+x) = xyz\left[\frac{1}{x}+\frac{1}{y}+\frac{1}{z}-1\right]$

$= xyz\left[x^{-1}+y^{-1}+z^{-1}-1\right]$

Now, $\frac{x^{-1}+y^{-1}+z^{-1}}{3} \geq \left(\frac{x+y+z}{3}\right)^{-1}$

$\Rightarrow (x+y)(y+z)(z+x) \geq 8xyz \qquad .....(ii)$

$\Rightarrow (1+x)(1+y)(1+z) \geq 8.8xyz$

$\left(1+\frac{1}{x}\right)\left(1+\frac{1}{y}\right)\left(1+\frac{1}{z}\right) \geq 64$

68. $\left|\text{adj}\left(I-2A^{-1}\right)\right| = \left(\frac{|B|}{|A|}\right)^2 = \left(\frac{10}{8}\right)^2$

69. $\frac{\cos^3\theta}{9a} = \frac{\sin^3\theta}{5b} = \lambda^3$

$\frac{9a}{\cos\theta} + \frac{5b}{\sin\theta} = 56$

$\frac{9a}{\lambda(9a)^{\frac{1}{3}}} + \frac{5b}{\lambda(5b)^{\frac{1}{3}}} = 56$

$\left[(9a)^{\frac{2}{3}} + (5b)^{\frac{2}{3}}\right]^3 = (56)^2$